

Vijay Baskar

Lead Software Engineer | AI/ML Developer | Mobile Systems & Scaled Engineering
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SUMMARY

Lead Software Engineer with 13+ years shipping Android, backend, and on-device AI/ML systems. Hands-on with Kotlin, Java, Jetpack Compose, MVVM, REST, Spring Boot, Firebase, TensorFlow, PyTorch, TensorFlow Lite, RAG, LangChain, and LLMs. Built production AI for self-checkout, voice assist, recipe generation, chatbots, and indoor navigation. Seeking AI, mobile, and backend engineering roles.

KEY ACHIEVEMENTS

- Saved over \$2M with AI-powered indoor navigation / product finder, replacing thousands of Bluetooth beacons.
- Reduced ANR crashes by 70% through Android architecture optimization; improved app performance by 30%.
- Designed Spring Boot microservices handling 500K+ users and reduced backend response time by 40%.
- Shipped AI agents and chatbots (Lisa, Voice Assist), AI recipe generation, object detection, recommendation engines, and on-device predictive text.

SKILLS

Programming Languages: Java, Kotlin, Dart, Python, SQL, MATLAB

Mobile: Kotlin, Java, Android SDK, Jetpack Compose, Jetpack (Room, Hilt, Navigation, WorkManager, DataStore), Kotlin Coroutines, Flow, MVVM, Clean Architecture, Material Design 3, iOS Interop, Flutter, Dart, Kotlin Multiplatform (KMP), BLoC, Provider, MVP
AI / ML: TensorFlow, TensorFlow Lite, PyTorch, Keras, ONNX, Model Quantization, Hugging Face Transformers, Scikit-learn, NumPy, Pandas, OpenCV, NLTK, on-device ML, edge ML

Generative AI: LLMs (GPT-4, Claude, Gemini, Llama 2/3), Prompt Engineering, RAG, LangChain, LlamaIndex, AI Agents, AI Chatbots, tool-calling

AI Shipped: Sentiment Analysis, Language Modeling, Object Detection, Recommendation Engines, Predictive Text, Voice Assist, AI Recipe Generation, ML-based Indoor Navigation

Backend & APIs: REST / RESTful APIs, GraphQL, WebSockets, Spring Boot, microservices, PostgreSQL, SQLite, Firebase (Auth, Firestore, FCM, Analytics), AWS, Vector Databases (Pinecone, FAISS, Weaviate, Chroma)

DevOps & Leadership: Jenkins, Fastlane, Bitrise, GitHub Actions, GitLab CI, Docker, Git, GitHub, Bitbucket, GitLab, Agile, Design Reviews, Mentoring, JUnit, Espresso, Flutter Test, CI/CD

Retail / IoT: ezyCart, self-checkout kiosk, NFC / Tap to Pay, Product Finder, Stock Audit, ezyList, Flutter SDK lite, Arduino, Raspberry Pi, Orange Pi, Bluetooth Beacons, IMU Sensors

EXPERIENCE

Retailtics SDN BHD — Lead Software Engineer | February 2021 – Present

Kuala Lumpur, Malaysia

- Spearheaded the AI-powered self-checkout smart trolley (ezyCart) on Android: system architecture, stakeholder design, and a real-time socket pipeline between IoT devices and the ezyCart Android app (Kotlin, WebSockets) for fast data sync without redundant permission requests.
- Shipped the self-checkout kiosk on the same stack (scan, bag, pay) and integrated payment gateways including Tap to Pay and NFC so checkout works on cart and kiosk.
- Built an on-cart AI recipe generation system: cuisine suggestions from basket items, cooking video, missing-product add-to-cart, plus in-aisle upsell and cross-sell ads.
- Delivered ML-based indoor navigation and product finder (search a SKU, navigate the aisle), replacing thousands of Bluetooth beacons and saving over \$2M.
- Shipped stock audit on the smart cart, Voice Assist AI for product and promo queries, and monitoring/troubleshooting for every active cart and self-checkout station.
- Designed and developed a cross-platform Flutter support-staff app, an Android TV cart monitor, and a Flutter SDK lite of ezyCart for scan-and-pay in vendor phone and tablet apps.
- Shipped ezyList (shared lists, nearby store price/availability, navigate, one-scan purchase, digital receipt, member promos) and Lisa, an AI shopping chatbot/assistant for live queries, recommendations, and promotions.
- Developed and deployed machine learning models (object detection, recommendation engines, on-device predictive text); led automated build pipeline and deployment across merchants and outlets; led split testing initiatives that drove additional revenue.
- Optimized high-performance Android architecture (70% fewer ANRs), Spring Boot microservices for 500K+ users (40% faster responses), and database access (60% faster queries via indexing, caching, and pagination). Integrated on-device TensorFlow Lite models with model quantization (~30% faster inference).

Colan infotech PVT LTD — Senior Software Engineer | November 2012 – February 2021

Chennai, India

- Delivered 20+ production mobile apps (Android/iOS): native Kotlin/Java (Android SDK, MVVM, Jetpack Compose) and Flutter (state management, widgets, animations, integrations).
- Implemented CI/CD with Jenkins, GitLab, and Bitrise; architected scalable mobile backends using AWS, Firebase, and RESTful APIs.

Advanced AI/ML Engineering Project — Proprietary R&D · Forex market indicator

- Designed and trained models that read FX market structure and render an inspectable indicator overlay for research and analysis. Applied time-series and deep-learning methods (including on-device and quantized inference patterns used in production mobile work) in a sequential build: indicator, then a calibrated signal layer, then an execution prototype with explicit risk limits.

EDUCATION

Anna University / GGR College of Engineering — Bachelor of Engineering in Computer Science | Vellore, India | June 2007 – November 2011

Target Software Solution — Mobile Application Development using Android and iOS | Chennai, India | January 2012 – October 2012

Additional coursework: SOLID Principles — Coursera; AI/ML Architecture — IBM; Master Prompting — Claude and Coursera